

Sección 2

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1. A die has two of its sides painted red, two black, and two yellow. If the die is rolled twice, describe a two-dimensional sample space for the experiment. What probabilities would you assign to the various sample points?

$$N_{\text{say}} = 1$$

$$N = 9$$

$$P = 1/9$$

3. A box contains 3 white and 2 black balls. Two balls are to be drawn from the box. Describe a two-dimensional sample space and assign probabilities to the sample points when (a) the balls are also numbered, (b) like colored balls cannot be distinguished.

a) Tenemos la posibilidad con 2 o puntos y con 1 dandolos

$$P = 1/10 \text{ para 2 puntos}$$

$$P = 3/10 \text{ para 1 punto.}$$

5. If the die of problem 1 is rolled until a red side comes up, describe a sample space for the experiment. What probabilities would you assign to the various sample points?

Las probabilidades serían:

$$P = 1/3, 2/3 \cdot 1/3, (2/3)^2 \cdot 1/3, \dots$$

7. One urn contains 2 white and 2 black balls; a second urn contains 2 white and 3 black balls. (a) If one ball is chosen from each urn, what is the probability that they will be the same color? (b) If an urn is selected by chance and one ball drawn

a) $N_{fav} = 1$ $N = 2$ $\Rightarrow P = \frac{1}{2}$

b) $N_{fav} = 1$ $N = 20$ $\Rightarrow P = \frac{1}{20}$

c) $N_{fav} = 11$ $N = 30$ $\Rightarrow P = \frac{11}{30}$

9. Assuming that the ratio of male children is $\frac{1}{2}$, find the probability that in a family of 5 children (a) all children will be of the same sex, (b) the three oldest will be boys and the two youngest will be girls, (c) three of them will be boys and two will be girls.

a) $P = \frac{2}{2^5} = \frac{2}{32} = \frac{1}{16}$

b) $P = \frac{1}{2^5} = \frac{1}{32}$

c) $P = \frac{10}{2^5} = \frac{10}{32} = \frac{5}{16}$

10. Successive drawings of a card from an ordinary deck are made with replacement each time. How many drawings are necessary before the probability is at least .9 that a spade will be obtained at least once?

$N_{fav} = 6!$ $N = 6^6$ $\Rightarrow P = \frac{N_{fav}}{N} = \frac{6!}{6^6} = \frac{720}{46656} = \frac{5}{324}$

least 12 that a space will be obtained at least once.

11. A box contains two honest coins and one coin that has two heads. A coin is selected at random from the box and tossed twice. (a) What is the probability it will show two heads? (b) If it shows three heads, what is the probability that it is the dishonest coin?

12. Two boxes contain 1 black and 3 white balls, and 3 black and 3 white balls.

$$a) N_{\text{fav}} = 2 \Rightarrow P = \frac{2}{64} = \frac{1}{32}$$

$$N = 64$$

$$b) N_{\text{fav}} = 1 \Rightarrow P = \frac{1}{64}$$

$$N = 64$$

$$c) N_{\text{fav}} = \binom{6}{3}$$

$$N = 64$$

$$\Rightarrow P = \frac{20}{64} = \frac{5}{16}$$

13. A coin is tossed once. If it comes up heads a die is rolled and you are paid the number showing in dollars. If it comes up tails two dice are rolled and you are paid in dollars the sum of the numbers showing. What is the probability that you will be paid at most 5 dollars?

$$N_{\text{fav}} = 10 \Rightarrow P = \frac{10}{18} = \frac{5}{9}$$

$$N = 18$$

15. Fourteen quarters and one five-dollar gold piece are in one purse and 15 quarters are in another purse. Five coins are taken from the first purse and placed in the second, and then 5 coins are taken from the second and placed in the first. What is the probability that after these transactions the gold coin will be found in the first purse?

$$N_{\text{fav}} = 3 \cdot (3)! = 18 \Rightarrow P = \frac{18}{24} = \frac{3}{4}$$

$$N = 4! = 24$$

17. Two dice are rolled. What is the probability that the sum of the faces exceeds 8, given that one or more of the faces is a 6?

$$N = 6^2 = 36$$

$$N_{\text{sum}} = 4$$

$$\Rightarrow P = \frac{4}{36} = \frac{1}{9}$$

$$\{(6, 3), (6, 4), (6, 5), (6, 6)\}$$

19. A bag contains 2 black balls and 2 white balls. A ball is drawn and replaced by a ball of the opposite color. Then another ball is drawn from the bag. Find the conditional probability that the first ball drawn was white, given that the second ball drawn was white.

$$P[b|w] = \frac{P[b \cap w]}{P[w]} = \frac{\frac{1}{8}}{\frac{1}{2}} = \frac{2}{8} = \frac{1}{4}$$

$$P[b] = \frac{1}{2} \quad P[b \cap b] = \frac{1}{8}$$

42. Two dice are rolled. Let X be the difference of the face numbers showing, the higher minus the lower, and 0 for ties. Find the density function of X .

Sean los dados con n caras, tal que:

$$X = |C_1 - C_2|$$

Donde $C_1, C_2 \in \{1, 2, 3, \dots, n\}$

Luego entonces si $C_1 = C_2$, $X = 0$ y si $C_1 \neq C_2$ lo más que puede valer X es $X = n - 1$.

$$\Rightarrow X \in [0, n-1]$$

Nuestro espacio de muestra se determina por los resultados de C_1 y C_2 , considerando que $(C_1, C_2) = (C_2, C_1)$

$$\Rightarrow N = \binom{n+1}{2} = \frac{n(n+1)}{2} = \frac{n^2+n}{2}$$

Luego vemos casos favorables hay en el valor de X , sea $X = 0$ cuantos existen.

$$\{1, 2, 3, \dots, n\}$$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $X_1 \quad X_2 \quad X_3 \quad X_n$
 $(1,1) \quad (2,2) \quad (3,3) \quad (n,n)$

Vemos que para $X = 0$ existen n casos favorables.

Para $X = 1$

$$\{ \underbrace{1, 2, 3, 4, \dots, n-1, n}_{\substack{X_1 \quad X_2 \quad X_3 \quad \dots \quad X_{n-1} \\ (1,2) \quad (2,3) \quad (3,4) \quad \dots \quad (n-1,n) \\ (2,1) \quad (3,2) \quad (4,3) \quad \dots \quad (n,n-1)}}$$

Vemos que como X toma un espacio de C_1 a C_2 , la cantidad de casos favorables están dados por el espacio reducido por la distancia X . Esto es:

$$N_f = n - x ; \forall x \in [0, n-1]$$

$$f(x) = \frac{N_f}{N} = \frac{n-x}{\frac{n^2+n}{2}} = \frac{2(n-x)}{n^2+n}$$

43. A box contains 4 red and 2 black balls. Two balls are drawn. Let X be the number of red balls obtained. Find the density function of X and also its distribution function.

Analizemos los casos favorables N_f y posibles N de X

Sea $X=0$

$$N_f = 2 \cdot 1$$

$X=1$

$$N_f = 4 \cdot 2 + 2 \cdot 4 \quad N = 6 \cdot 5$$

$X=2$

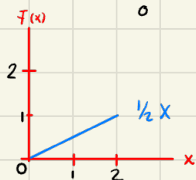
$$N_f = 4 \cdot 3$$

$$\Rightarrow \text{Para este caso podemos decir que } f(x) = \begin{cases} \frac{1}{15} & ; x=0 \\ \frac{8}{15} & ; x=1 \\ \frac{6}{15} & ; x=2 \end{cases}$$
$$\Rightarrow F(0) = \frac{1}{15} ; F(1) = \frac{9}{15} ; f(2) = 1$$

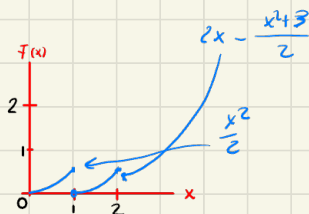
65. Find the distribution function $F(x)$ and graph it if the density of X is (a) $f(x) = \frac{1}{2}$, $0 \leq x \leq 2$, (b) $f(x) = x$ for $0 \leq x \leq 1$ and $f(x) = -x + 2$ for $1 < x \leq 2$, (c) $f(x) = [\pi(1 + x^2)]^{-1}$.

Supongamos $\alpha = X$

a) $F(\alpha) = \int_0^{\alpha} \frac{1}{2} dx = \frac{1}{2} [x]_0^{\alpha} = \frac{1}{2} \alpha = \frac{1}{2} X$



b) $f(x) = \begin{cases} x & ; 0 \leq x \leq 1 \\ 2-x & ; 1 < x \leq 2 \end{cases} \Rightarrow F(\alpha) = \int_0^{\alpha} f(x) dx = \begin{cases} \int_0^{\alpha} x dx & ; 0 \leq x \leq 1 \\ \int_1^{\alpha} 2-x dx & ; 1 < x \leq 2 \end{cases} = \begin{cases} \frac{\alpha^2}{2} \\ 2\alpha - \frac{\alpha^2}{2} - [2 - \frac{1}{2}] \end{cases} = \begin{cases} \frac{\alpha^2}{2} \\ 2\alpha - \frac{\alpha^2 + 3}{2} \end{cases} = \begin{cases} \frac{x^2}{2} & ; 0 \leq x \leq 1 \\ 2x - \frac{x^2 + 3}{2} & ; 1 < x \leq 2 \end{cases}$



c) $f(x) = \frac{1}{\pi(1+x^2)} \Rightarrow F(\alpha) = \int_{-\infty}^{\alpha} \frac{dx}{\pi(1+x^2)} = \frac{1}{\pi} \int \frac{dx}{(1+x^2)} = \frac{1}{\pi} [\arctan(x)]_{-\infty}^{\alpha} = \frac{1}{\pi} [\arctan(\alpha) - \arctan(-\infty)]$

$F(x) = \frac{1}{\pi} [\arctan(x) + \arctan(\infty)] = \frac{1}{\pi} [\arctan(x) + \frac{\pi}{2}] = \frac{1}{\pi} \arctan(x) + \frac{1}{2} ; x \in (-\infty, \infty) \text{ y } F(x) \in [0,1]$

